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Research Paper

Leveraging multisource data for off-street parking occupancy prediction: a twofold analysis

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ABSTRACT

Predictive parking management may have far-reaching implications on network traffic and emissions, but requires high-performance models for predicting parking demand and supply. This paper focuses on developing machine learning (ML) models for accurate network level short-term and long-term off-street parking occupancy prediction for multiple facilities at once. The models were trained using a parking occupancy dataset from four central parking facilities in Athens, Greece, also fed with information about traffic evolution, transit operations, and weather. From the various modeling attempts conducted, the long short-term memory (LSTM) models with attention mechanism achieved the best performance in short-term forecasting while the transformer-based models excelled in long-term predictions. Additionally, Shapley additive explanations (SHAP) values and permutation importance (PI) were computed for all models to provide interpretability by identifying the most influential features, such as loop detectors in arterials leading to the city center and central metro stations, in shaping the model's predictions. The further analysis of the attention weights indicates that recent past steps play an influential role in short-term forecasting, while long-term predictions show several peaks indicating that they benefit from different time points across the input sequence. Findings of our work can provide applicable solutions for optimizing parking management, enabling demand-responsive strategies, and providing information to users.

1 Introduction

Parking management is a critical component of urban planning as it directly affects traffic flow, environmental sustainability, and the economy of city centers (Shoup, 2006). The design and implementation of dynamic demand-responsive parking management strategies, referring either to on-street or off-street parking spaces or both, including parking space reallocation and dynamic pricing, may have significant impacts on the overall efficiency of the network and enhance users' level of service. To this direction, many cities and parking facilities have adopted demand responsive strategies to control occupancy of parking spaces and redirect users to underutilized parking areas. Some prominent examples are the SFPark demand-responsive parking pricing system implemented in San Francisco (Pierce and Shoup, 2018) and the Los Angeles

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ExpressPark (Ghent, 2015). Other cities that have adopted similar techniques are Seattle in the USA and Madrid in Europe (Friesen and Mingardo, 2020; Ottosson et al., 2013).

The success of these strategies depends on the prediction of parking availability or parking occupancy. Parking occupancy is a crucial piece of information as it affects cruising times and the implementation of optimal pricing strategies (Tamrazian et al., 2015). The prediction of parking availability, combined with modern data collection and processing capabilities, has been recognized as a challenge since the early 1990s (Bonsall, 1991). However, recent advancements, such as Internet of Things (IoT), sensor technology, and big data analytics, made the practical application and deployment of these predictive parking strategies feasible (Mantouka et al., 2021; Vlahogianni et al., 2016). More precisely, parking occupancy predictions have been used in many state-of-the-art agent-based and event-based simulations in order to enhance decision-making processes related to parking management. Mei et al. (2020) developed an agent-based simulation model that represents the coexistence of reservation and non-reservation parking vehicles, utilizing the predicted occupancy rates of specific parking lots at anticipated vehicle arrival times. Nugraha and Tanamas (2017) proposed an event-driven algorithm to dynamically allocate off-street parking spaces. Their goal was to effectively reduce the time drivers spend searching for parking and maintain high utilization levels through real-time adjustments to reservations based on vehicle arrivals. In that scheme, occupancy predictions could provide foresight into future demand patterns, allowing the event-driven simulation to anticipate peaks in parking demand or identify periods of low occupancy in advance. With this predictive approach, the algorithm could dynamically adjust the allocation of parking spaces or reservations based on expected vehicle arrivals, rather than reacting solely to real-time conditions.

There are two main approaches to parking availability and occupancy prediction (Xiao et al., 2018). The first one considers stochastic arrival and departure processes, often employing queuing theory to predict the flow and accumulation of parked vehicles. Typically, the process of vehicles arriving at a parking facility is modeled as a single queue following Poisson distribution (Caliskan et al., 2007). The parking duration is often modeled using either negative exponential distribution (Millard-Ball et al., 2014) or discretized Gamma distribution (Caicedo, 2009; Caicedo et al., 2012). Despite their theoretical interest, these earlier studies often fail to validate their models in real-world data.

The second approach applies machine learning (ML) techniques to predict future parking availability straight from historical data. Most approaches differ from each other in the way they handle data and the models they use (Awan et al., 2020). Zhao et al. (2020) compared four classic ML models, and found that combining weekends and weekdays in a single dataset was more effective for predicting large-scale parking lot occupancy, while separate forecasting was better for medium-scale lots . Jelen et al. (2021) tested ensemble methods that were trained with data coming from IoT-equipped smart parking facilities. Tekouabou et al. (2022) used a similar approach to test random forests and CatBoost in a dataset incorporating weather and event data. The temporal patterns of parking occupancy have been also widely investigated, using time-series and deep learning models like long short-term memory (LSTM) (Li et al., 2018), recurrent neural networks (RNNs) (Camero et al., 2019), or convolutional neural networks (CNNs), where image time-series were available (Amato et al., 2017).

Recent studies have demonstrated the efficacy of hybrid models in capturing a wider range of patterns in parking occupancy. Zhang et al. (2021) developed a PewLSTM to forecast parking behavior by integrating weather conditions and periodic parking patterns. To handle the noise in the data and to also consider the spatial correlation of vacant parking spaces in multiple parking facilities, Zeng et al. (2022) used a combination of wavelet transform and bidirectional LSTMs. Similarly, Gao et al. (2023) integrated a discrete wavelet transform for data denoising, convolutional gated recurrent units for capturing the spatiotemporal correlations of different parking facilities, and a batch normalization-rectified linear activation-convolution composite function for feature enhancement. Mufida et al. (2023) studied the spatial and temporal correlations of parking lots with a k-means clustering model to cluster parking lots based on their spatial profiles and a dynamic time warping (DTW) distance matrix for the temporal clustering. After the clustering, a single model for one of the parking facilities was trained and adapted for the other parking lots in the same cluster, minimizing the computational cost of training separate models.

Nevertheless, the correlation between parking dynamics and broader traffic metrics remains a rather underexplored area. Only a few studies have attempted to model the effects of traffic on parking occupancy and vice versa. Ma et al. (2017) explored how parking garage occupancy levels influence traffic speeds, highlighting that congestion tends to build up more rapidly during the morning peak hours as parking lots reach certain occupancy levels. Pi et al. (2019) integrated parking into a multi-modal dynamic user equilibrium model, revealing how parking costs and the availability of transit options influence both individual costs and overall system efficiency. Additionally, the effects of smart parking systems show that intelligent parking solutions can lead to reduced traffic congestion and thus improve transit usage patterns (Krishnamurthy and Ngo, 2020). Moreover, a comparative analysis of parking demand considering transit service levels was conducted, showcasing that higher transit service rates reduce parking demand (Rowe et al., 2011). Interestingly, few studies that developed prediction models, such as the ones discussed in the previous paragraph, have also incorporated traffic variables, such as speed and volume into their models (Badii et al., 2018; Provoost et al., 2020; Yang et al., 2019).

Despite the significant advancements in parking occupancy prediction models, several gaps remain that limit the applicability of the developed approaches in real world settings and the efficiency of the existing parking management systems. One key area that requires further exploration is the explainability of parking occupancy predictions, as current models often function as black boxes, providing little insight into the factors that affect their forecasts. Explainability could be a useful tool for parking management and enhance informed decision making both for traffic managers and the end users. The understanding of the parameters that affect parking demand is vital for the design and application of efficient parking management strategies with their resulting benefits that include reduced traffic congestion and road safety, improved environmental conditions, and driving experience. Additionally, there is a need for models that support proactive management by predicting occupancy several hours ahead for multiple parking locations, allowing for information provision to the drivers. Most existing approaches are limited to short-term forecasting, and mostly focus on a single parking facility, while parking demand extends over a large part of the network with unbalanced characteristics contrary to the supply side which is provided at specific points with limited capacity. Finally, the integration of multimodal datasets, including traffic volumes and public transit usage, is currently underutilized. Searching for parking is a significant part of the traffic, and therefore the consideration of traffic patterns in the analysis of parking can highlight critical relationships between traffic conditions and indicators. This integration is crucial for understanding parking dynamics within the transportation network and improving occupancy predictions.

The aim of this paper is to develop accurate models for parking occupancy prediction for several parking facilities simultaneously using a parking dataset containing occupancy data from four parking facilities of Athens, Greece, enhanced with traffic, transit, and weather data from the same region in two distinct levels. The first level focuses on off-street parking occupancy predictions using a classic and explainable ML technique. The second level extends the analysis by leveraging advanced sequential models to capture the temporal dynamics and long-term dependencies in parking occupancy. This includes the development of LSTM models with an attention mechanism and transformer-based models to explore and compare their effectiveness. Additionally, a simple 'reference day' baseline method is introduced to compare the performance of these sophisticated models against established, simpler approaches in transportation science. The goal is to investigate which methods result in more accurate parking occupancy predictions and which external multi-source data can enhance them. This includes an integration of traffic volumes and public transit ticket validations to account for a multimodal approach. This is one among very limited studies where an extensive integration of diverse mobility data streams is incorporated into parking occupancy prediction.

The remainder of the paper is organized as follows. Section 2 presents the methodology and models used. Section 3, an overview of the datasets and explanatory data analysis is provided. In Section 4, the implementation and evaluation of the methods are presented, and the main findings are discussed. Finally, in Section 5, the main conclusions and future research directions are given.

2 Methodological approach

2.1 Problem formulation

To address the problem of occupancy prediction for proactive parking management, two distinct approaches were considered: predictive modeling using ML and deep learning time-series forecasting. In the first part of the analysis, parking occupancy is predicted using a classic ML approach with the XGBoost regressor, leveraging a multimodal dataset with autoregressive values. The dataset incorporates parking occupancy, traffic volumes, public transport ticket validations, weather conditions, and temporal metrics. XGBoost was chosen as it is a robust and easily explainable model. Two interpretability metrics, permutation importance (PI) and Shapley additive explanations (SHAP), were employed to assess variable importance. The main objective was to develop a model to accurately predict parking occupancy by maximizing the use of widely available data sources.

The second part of the analysis focused on the temporal nature of the parking occupancy prediction problem and the relationship between parking occupancy and the road network dynamics (traffic volumes and transit usage). This is achieved by creating and comparing two advanced deep learning models capable of forecasting the occupancy rates of four parking lots at different time horizons. The first one is an LSTM neural network enhanced with an attention layer, and the second one is a state-of-the-art transformer based architecture for time series forecasting. Both models can capture long-term and short-term dependencies in sequential data, while the attention layers improve accuracy by highlighting the most relevant parts of the input sequences. Unlike simpler RNNs, which can face challenges with longer sequences due to vanishing gradients, LSTMs and transformers address these limitations more effectively, thereby balancing model complexity and predictive power. In addition, SHAP values and PI were also computed for the best sequential model to evaluate feature importance and interpretability.

Due to the large dataset for the time series forecasting models, feature selection was performed using mutual information (MI). Additionally, as a baseline comparison, a "reference day" approach was implemented. This approach involves predicting the parking occupancy at a given time by assuming it will be equal to the occupancy observed at the same time one week earlier (on the same weekday, accounting for holidays). By incorporating this basic yet reliable technique, we aim to benchmark the performance of our advanced models against a simple, transparent method to better evaluate the effectiveness of our ML and deep learning approaches. Lastly, to investigate the contribution of the multimodal variables to more accurate predictions, a model is also developed using only previous parking occupancies as input variables.

This twofold approach provides a framework for analyzing both external features and dynamic temporal relationships, offering insights into the factors that influence parking occupancy rates while also enabling highly accurate forecasting for proactive parking management.

2.2 Theoretical background

2.2.1 XGBoost model

XGBoost is an ML algorithm based on the boosting technique. Boosting refers to any ensemble method that can combine several weak learners into a strong learner. In the case of XGBoost, the weak learners are individual decision trees. Initially, a model is trained on a subset of training data. With this model, predictions are made on the entire training dataset, and the error is calculated. Usually, these predictions are weak, and the model needs to be strengthened in next iterations by fitting into the residuals of the previous step, instead of changing the weights of the samples that were not predicted correctly.

Let $F = \{f_1, f_2, f_3, \cdots, f_m\}$ represent the set of m weak learners, where each f_t is a decision tree. The final prediction for an instance x_i is calculated by summing the outputs of each weak learner as follows:

$$\widehat{y}_i = \sum_{t=1}^{m} f_t(x_i),\ f_t \in F. \tag{1}$$

2.2.2. Interpretability metrics

XGBoost is an easily explainable model. Therefore, to investigate the feature importance, two types of measures are implemented. The first one is the PI, and the second are the SHAP values. More specifically, PI, is a model-agnostic technique to assess the importance of individual features in an ML model. The PI of each feature is estimated as the decrease of the model's accuracy when randomly shuffling the specific feature's values for a predefined number of iterations. Therefore, it is a metric of each feature's importance for estimating the dependent variable's value accurately. PI (I_{P}) is given by

$$I_{\mathrm{P}j} = s - \frac{1}{K}\sum_{k=1}^{K} s_{k,j}, \tag{2}$$

where j is a feature of the model, s is the accuracy, K is the number of iterations and $s_{k,j}$ is the accuracy of the k^{th} iteration after permuting feature j's values.

SHAP values provide an explanation of any ML model by assigning each feature a contribution to the model's prediction. They are based on cooperative game theory and the Shapley value, which was originally developed to fairly distribute payoffs in multi-agent systems. SHAP values in ML are calculated by systematically altering input features and observing how these changes affect the model's predictions. Specifically, the SHAP value represents the average expected marginal contribution of an input feature after considering all possible combinations.

Let $f(x)$ be the model's prediction for an instance x, and let $S \subseteq N$ be any subset of features, where N is the full set of features. The Shapley value φ for a feature j is given by

$$\phi_j = \sum\nolimits_{S \subseteq N\{j\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!}[f(S \cup \{j\}) - f(S)], \tag{3}$$

where S is a subset of the features excluding j, $f(S)$ is the model's prediction using the features in subset S, $f(S \cup \{j\})$ is the model's prediction when feature j is added to subset S, and $|N|$ is the total number of features.

Both metrics are used in this study as a way to provide complementary insights into feature importance. While PI offers a straightforward approach to calculate the overall impact of features on model performance through model accuracy, SHAP values provide an insight into the individual contribution of each feature, considering their interaction with other features. Also, metric like these hold considerable managerial implication (Fafoutellis et al., 2022). By combining both approaches, transportation engineers can obtain a comprehensive understanding of how various factors affect parking occupancy predictions. For example, while PI might highlight that weather conditions like rainfall significantly impact overall model accuracy, SHAP values can reveal the extent to which rainfall specifically affects occupancy at different times of the day or under various circumstances. This dual approach enables engineers to develop more effective and targeted strategies for managing parking resources.

2.2.3 LSTM model

The proposed model architecture consists of an LSTM network with an attention layer. This combination leverages the strength of LSTM in handling sequential data and the attention mechanism to focus on important parts of the input sequence. The architecture processes multiple time series in parallel within a single model framework.

The LSTM network is designed to handle long-term and short-term dependencies in sequential data (Hochreiter and Schmidhuber, 1997). LSTM is a type of RNN that includes special units called LSTM blocks, composed of input gates, forget gates, and output gates. These gates capture the complex non-linear relationships among different factors. The LSTM gate operates with Eqs. (4)–(8):

$$\text{Forget gate}: F_t = \sigma(W_f \cdot [H_{t-1}, X_t] + b_f. \tag{4}$$

$$\text{Input gate}: I_t = \sigma(W_i \cdot [H_{t-1}, X_t] + b_i). \tag{5}$$

$$\text{Cell state}: \widetilde{C_t} = \tanh(W_C \cdot [H_{t-1}, X_t] + b_C), \tag{6}$$

$$C_t = F_t \cdot C_{t-1} + i_t \cdot \widetilde{C_t}. \tag{7}$$

$$\text{Hidden state}: H_t = O_t \cdot \tanh(C_t). \tag{8}$$

In the above equations, X_t is the input at time step t, which in our study is a multidimensional vector consisting of parking occupancy data, traffic volumes, public transit ticket validations, and weather conditions, H_{t-1} is the hidden state from the previous time step, σ denotes the sigmoid function, and tanh the hyperbolic tangent function. W_f, W_i, W_C, W_o are the weight matrices, and b_f, b_i, b_C, b_o are the bias vectors.

For a clearer understanding of how an LSTM block operates, Fig. 1 demonstrates the architecture and interactions between the gates within an LSTM block.

2.2.4 Attention mechanism

The attention mechanism allows the model to focus on different parts of the input sequence when making a prediction (Vaswani et al., 2017). The traditional LSTM network, despite its ability to capture long-term dependencies, treats all the past information with equal importance. This can be suboptimal for time series forecasting tasks where some parts of the sequence might be more relevant than others for predicting future values. The attention score determines the relevance of each hidden state at a given time step, and is calculated by

$$e_t = v_e^{\mathrm{T}} \cdot \tanh(W_e \cdot H_t + b_e), \tag{9}$$

where e_t represents the attention score at time step t, v_e is a learnable weight vector, W_e is the weight matrix which remains the same for all time steps t during the calculation of attention scores, H_t is the hidden state at time step t, and b_e is the bias term. This score quantifies the importance of the hidden state H_t, which represents the model's internal state at time step t. The attention weights, which normalize the attention scores to a probability distribution, are computed by

$$\alpha_t = \frac{\exp(e_t)}{\sum_{t'} \exp(e_{t'})}, \tag{10}$$

where the denominator is the sum of the exponentiated attention scores over all time steps t'. This is also known as Softmax function. Finally, the context vector, which captures the most relevant information from the input sequence, is calculated by

$$c_t = \sum\nolimits_t \alpha_t H_t. \tag{11}$$

The context vector essentially summarizes the input sequence, emphasizing the parts considered most important by the attention mechanism. Fig. 2 shows the structure of an LSTM with attention layer network.

2.2.5 Transformer model

The implemented transformer model is designed based on an architecture designed for sequence-to-sequence tasks, inspired by the original transformer model (Vaswani et al., 2017). The model employs multi-head self-attention layers to focus on the most relevant parts of the input sequence.

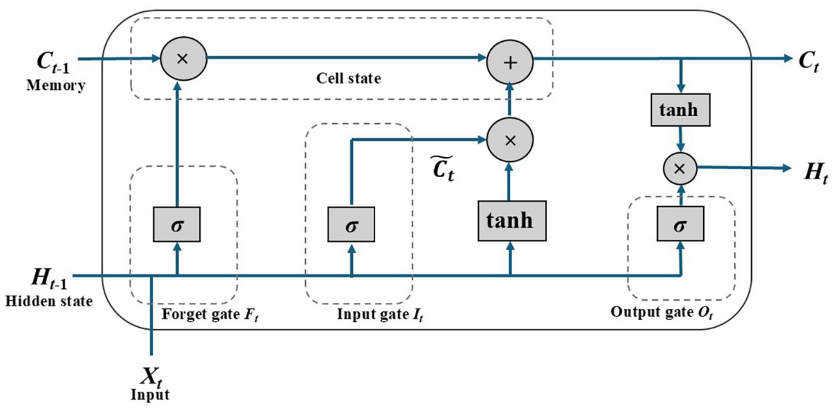

Fig. 1 Architecture of an LSTM block.

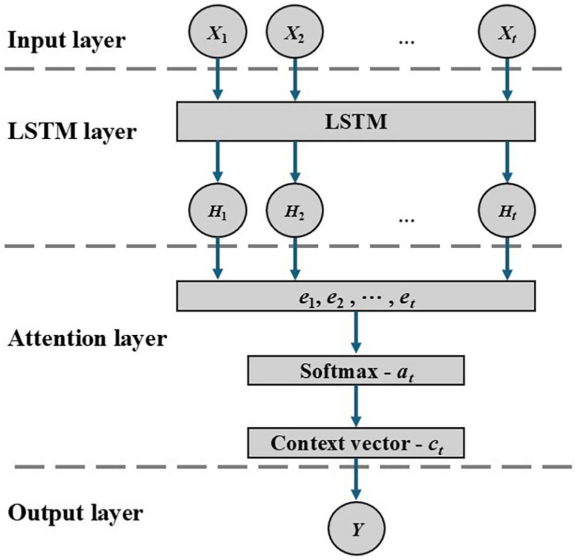

Fig. 2 LSTM with attention layer network architecture.

The core component of the transformer architecture is the self-attention mechanism, which allows each position in a sequence to attend to all other positions. In the context of time series data, this capability allows the model to capture both local and long-term dependencies across the input sequence. The self-attention mechanism is computed by projecting the input sequence into three different vectors: queries (Q), keys (K), and values (V). The output is then computed as a weighted sum of the values, where the weights are determined by the compatibility of queries with keys. This can be expressed by

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^{\mathrm{T}}}{\sqrt{d_k}}\right)V, \tag{12}$$

where d_{k} is the dimensionality of the keys, ensuring the dot products are appropriately scaled. In this case, multi-head attention is employed where multiple attention heads are applied in parallel, allowing the model to focus on different parts of the input sequence simultaneously. The outputs of these heads are concatenated and linearly transformed before being passed on to subsequent layers. Following this, a feed-forward network (FFN) is applied to each position in the sequence independently, allowing the model to transform each input step in a non-linear fashion. For a given input x, the output of the FFN is given by

$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2, \tag{13}$$

where W_1, W_2, b_1, and b_2 are learned parameters. To stabilize training, layer normalization is applied after the attention and FFN layers, helping the model learn effectively and maintain a stable distribution of activations.

For sequence-to-sequence tasks such as time series forecasting, where the output depends on both past and future states, the transformer model adopts an encoder-decoder architecture. The encoder processes the input sequence, and produces a set of encoded representations, while the decoder attends to these encoded representations through cross-attention. In cross-attention, the query comes from the decoder, while the keys and values come from the encoder. This mechanism allows the decoder to focus on relevant parts of the input sequence while generating the output sequence.

$$\text{Cross-Attention}(Q_{\text{decoder}}, K_{\text{encoder}}, V_{\text{encoder}}) = \text{softmax}\left(\frac{Q_{\text{decoder}}K_{\text{encoder}}^{\mathrm{T}}}{\sqrt{d_k}}\right)V_{\text{encoder}}. \tag{14}$$

The combination of self-attention, cross-attention, and feed-forward layers forms the backbone of the transformer model. However, unlike recurrent models that process inputs sequentially, the transformer lacks a built-in notion of order in the sequence. To address this, positional encoding is added to the input embeddings to provide the model with information about the relative positions of tokens in the sequence. Positional encoding (E) is defined as a sinusoidal function of the position (Pos) in the sequence, and is mathematically expressed by

$$E_{\text{pos},\,2i} = \sin\left(\frac{p_{\text{os}}}{10\,000^{\frac{2i}{d_{\text{model}}}}}\right), \tag{15}$$

$$E_{\mathrm{pos},\,2i+1} = \cos\left(\frac{p_{\mathrm{os}}}{10\,000^{\frac{2i}{d_{\mathrm{model}}}}}\right), \tag{16}$$

where p_{os} is the position in the sequence, and i is the dimension index. This encoding enables the model to discern the sequential order of inputs. Fig. 3 displays the layers and structure of the transformer model.

2.2.6 Evaluation metrics

In this section the key evaluation metrics used to quantify the accuracy and effectiveness of the developed models are presented.

- Mean squared error (MSE, e_{MSE}): MSE measures the average of the squares of the differences between the predicted and actual values.

$$e_{\mathrm{MSE}} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2. \tag{17}$$

- Mean absolute percentage error (MAPE, e_{MAPE}): MAPE provides the average absolute percentage error between predicted and actual values, offering an intuitive understanding of the model's prediction error as a percentage.

$$e_{\mathrm{MAPE}} = \frac{100}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right|. \tag{18}$$

- Mean absolute error (e_{MAE}): MAE measures the average absolute differences between the predicted and actual values. Unlike MSE, it does not square the errors, thus treating all deviations equally.

$$e_{\mathrm{MAE}} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \hat{y}_i\right|. \tag{19}$$

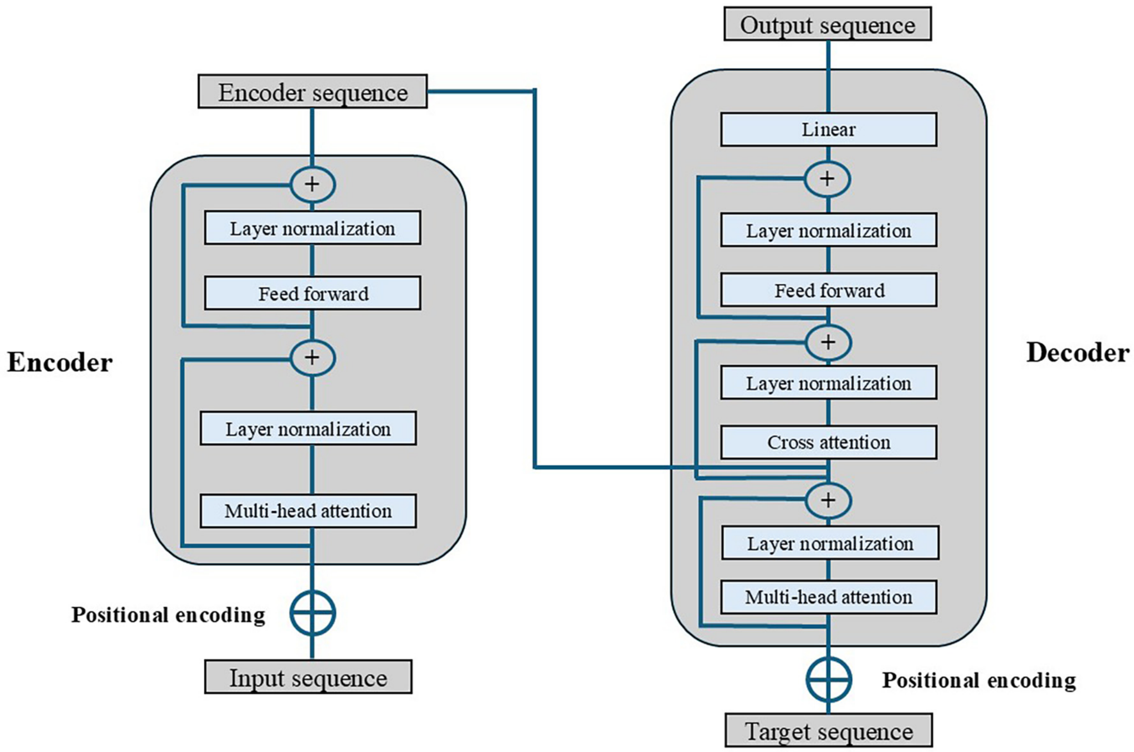

Fig. 3 Transformer model for time series forecasting architecture.

- Coefficient of determination (R^2): R^2 quantifies how well the model explains the variability in the actual data. It ranges from 0 to 1, where a value closer to 1 indicates better model performance. R^2 is calculated as:

$$R^2 = 1 - \frac{\sum_{i=1}^{n} \left(y_i - \widehat{y}_i\right)^2}{\sum_{i=1}^{n} \left(y_i - \bar{y}\right)^2}. \tag{20}$$

In these equations, y_i represents the actual value of the dependent variable for the i-th observation, $\widehat{y}_i$ is the predicted value, and n is the total number of observations, $\bar{y}$ denotes the mean of the actual values y_i.

3 Dataset description

In order to model parking characteristics, it is essential to study data covering a range of factors that could possibly affect parking decisions, such as traffic conditions, public transit usage, and weather patterns. Such datasets are often hard to obtain and analyze together due to the need to combine data from different sources. Also, parking data, either on-street or off-street, are not often open-source and readily available. Data used in this analysis spans from the 20th of March 2023 to the 30th of April 2023 and comes from four different sources: parking facilities, loop detectors, metro ticket validations, and weather datasets. The integration of all these diverse data sources into a single analysis enables a more accurate understanding of the problem of parking dynamics, and provides valuable insights that are typically not included in other state-of-the-art studies.

Parking data refer to occupancy rates recorded at hourly intervals for four central parking garages in Athens, Greece. Two of them are in the city center (abbreviated as EMB and KED), and the two others are located at entrances of the city center (abbreviated as PYR and OKN). Traffic data are sourced from loop detectors throughout the Athens network and provide hourly traffic volumes per lane. Transit data include ticket validations from all metro stations across the city. The spatial distribution of the dataset can be seen in Fig. 4, Additionally, weather data, gathered from NASA's weather portal, refer to temperature readings and rainfall information.

In order to provide a first glimpse at the data, Fig. 5 illustrates the hourly occupancy rates for the four parking facilities over the span of six weeks. It demonstrates that all the parking garages reach or nearly reach full capacity, with the EMB parking, which is the most central facility out of the four, often exceeding its capacity. That means that vehicles inside the garage are being double parked and actively managed by the parking facility's staff to prevent operational disruptions.

The data reveal strong daily patterns, with clear daily peaks during typical working hours (9.00–18.00) and lower occupancy rates during late evenings and early mornings. These periodic fluctuations suggest a high correlation with daily commuting patterns, likely influenced by the surrounding traffic and transportation network. This hypothesis is further investigated by comparing the time-series of the parking facilities occupancies with the traffic volumes recorded by their nearest loop detectors and the corresponding metro station's ticket validations.

For example, Fig. 6 illustrates the occupancy rates of the EMB parking garage alongside traffic volumes recorded by the nearest loop detector and with ticket validations at the nearest metro station. The data patterns suggest a correlation between these metrics, with peaks in traffic volume and validations often aligning with peaks in parking occupancy. This alignment suggests that increased traffic flow directly impacts the demand for parking spaces, particularly during the morning peak hours. Also, the multimodal nature of the problem is highlighted, where public transit and private vehicle usage are interconnected. For instance, during morning peak hours, increased transit validations indicate a shift towards public transport, yet a corresponding rise in parking occupancy suggests that a significant portion of the population is driving to transit hubs. The scatterplots further emphasize that positive correlation. The same patterns are observed for the other three parking facilities and their respective nearest loop detectors and metro stations.

3.1 Data preprocessing

The preprocessing of the data consisted of four steps. First, feature selection was performed using MI to identify the most correlated loop detectors and metro stations for the four parking facilities. MI is a metric able to capture both linear and nonlinear dependencies, essential for complex data such as traffic flow and metro station usage, where relationships with parking occupancy may not be purely linear. As a result, 37 loop detectors out of 389 available and 14 metro stations out of 62 available were selected for the analysis as can be seen in Fig. 7. The figure shows that the most correlated metro stations are those located nearest to the parking garages or the ones in the suburbs, while the loop detectors are situated along major arterial roads that lead to the city center. Second, autoregressive features were created for the XGBoost model. To account for the temporal nature of parking occupancy and to ensure a fair comparison with the rest of the developed models. Autoregressive values were generated by transforming the time-series data into lagged features. Specifically, each feature was augmented with its values from prior time steps, i.e., $X_t, X_{t-1}, \cdots, X_{t-w+1}$, where w represents the number of lagged time steps considered. In this case, w was set to 24 to incorporate a full day of historical data into the feature set. Additionally, the target variable (Y) was lagged to enable predictions several hours ahead. Third, a MinMax scaler was applied to scale the data

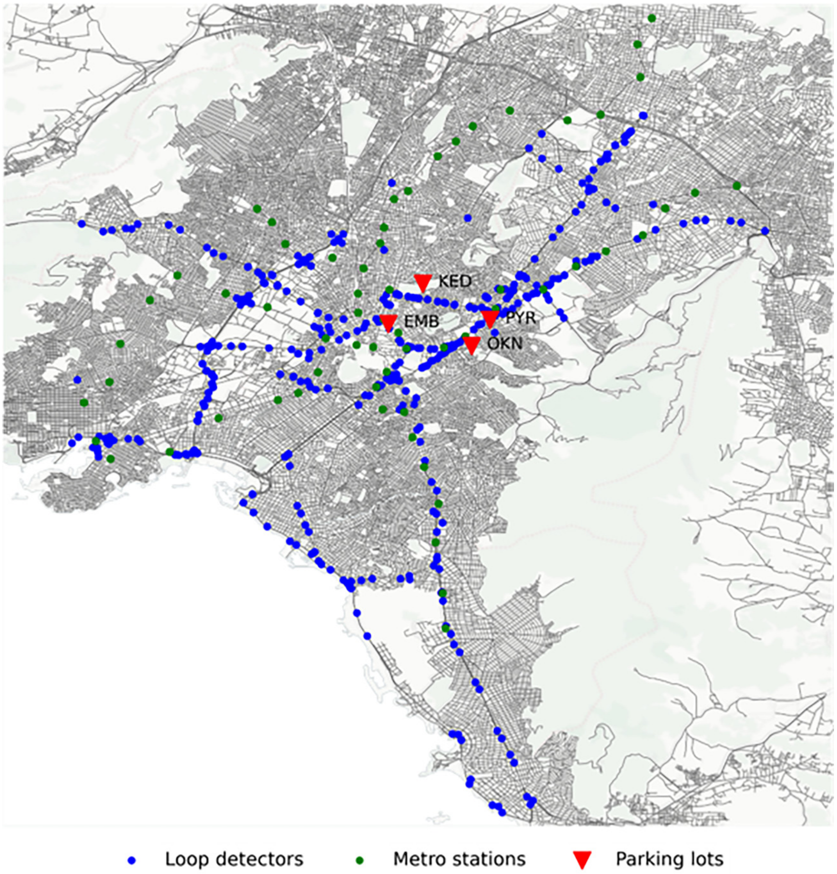

Fig. 4 Locations of loop detectors, metro stations, and central parking facilities used in the analysis.

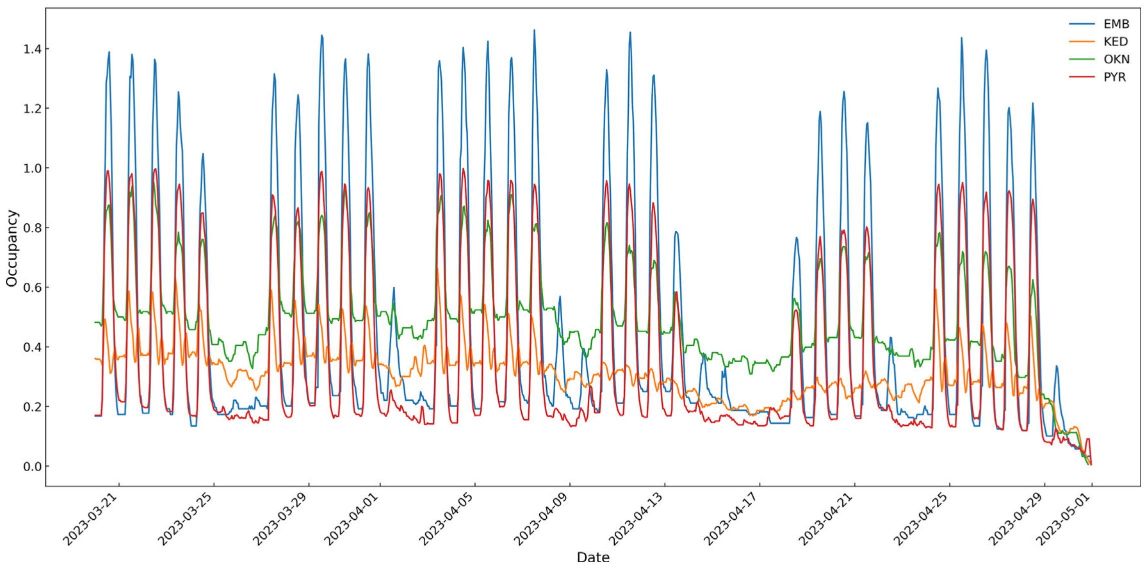

Fig. 5 Occupancy time-series of the four parking facilities.

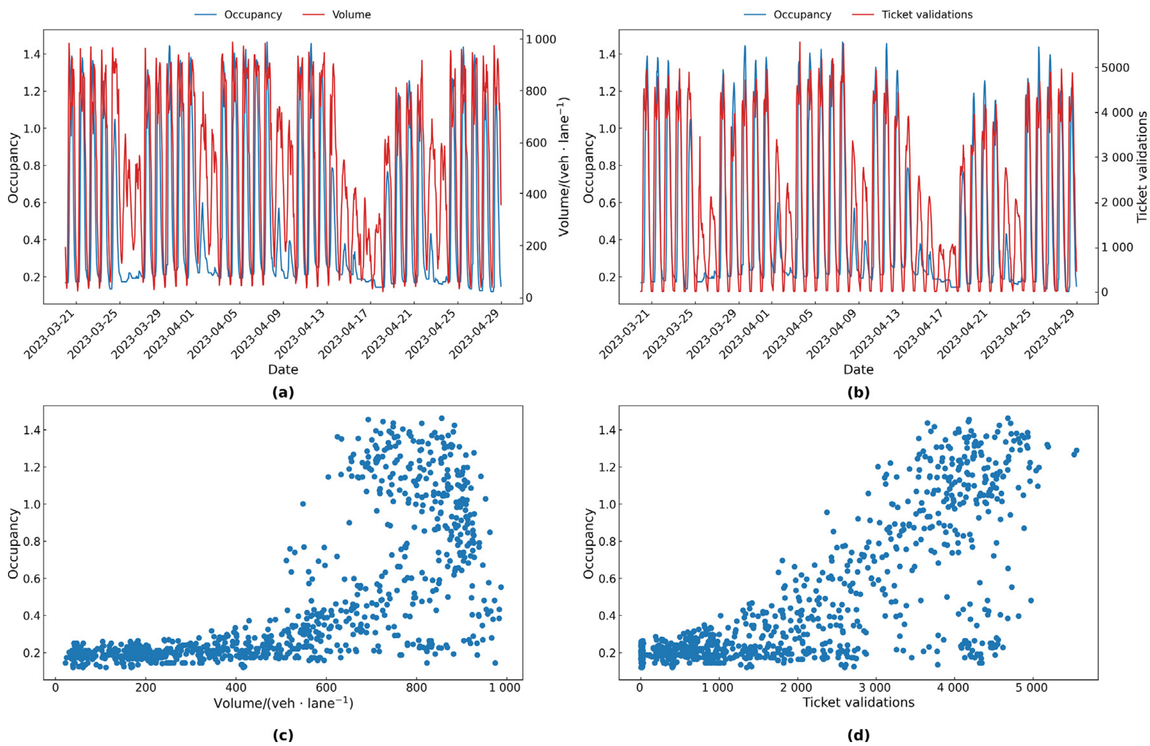

Fig. 6 EMB parking occupancy time-series compared with traffic volumes from the nearest loop detector (a) and with the nearest metro station's ticket validations (b) and their respective scatterplots (c), (d).

between 0 and 1, ensuring equal contribution of all features to the learning process. Lastly, a sliding window approach was adopted to transform the time-series data for the sequential models into sequences, with each input sequence represented as $\{X_{t-w+1}, X_{t-w+2}, \cdots, X_t\}$ and the corresponding output as Y_{t-1}. w also set to 24, ensuring consistency across the preprocessing pipeline and facilitating the sequential models' ability to capture temporal patterns over an entire day.

4 Implementation and results

4.1 Model training

The XGBoost model was trained using a multimodal dataset, incorporating both time-related variables (such as hour of the day, day of the week, and a binary variable indicating whether it is a weekend or not) and weather-related variables (such as temperature and rain), along with autoregressive features created by lagging parking occupancy, traffic volume, and transit data from the most correlated loop detectors and metro stations (Table 3). By incorporating autoregressive features, the model effectively accounted for temporal patterns without requiring sequential processing, making it an easily interpretable tool for parking occupancy prediction. The hyperparameters used in the training were derived through an extensive grid search and are as follows (Table 1).

- Learning rate: Set at 0.12, which controls the step size at each iteration while moving towards a minimum of the loss function.
- Number of estimators: Set at 70, which indicates the number of trees in the model.
- Maximum depth: Set at 6, which determines the maximum depth of a tree and help prevent overfitting by limiting the number of splits in a tree.
- Minimum child weight: Set at 1, which specifies the minimum sum of instance weight needed in a child.
- Column sample by tree: Set at 0.8, this parameter controls the fraction of features to be randomly sampled for each tree.

Subsample: Set at 0.9, which denotes the fraction of samples to be used for fitting the individual trees and helps to prevent overfitting by adding randomness to the training process.

The LSTM models use the past 24 h to predict the future occupancy rates at different time horizons. The input features used for the training are the selected features from the correlation analysis that include the volumes per lane of the loop detectors and the ticket validations of the metro stations. In addition to the transportation network data, the input time-

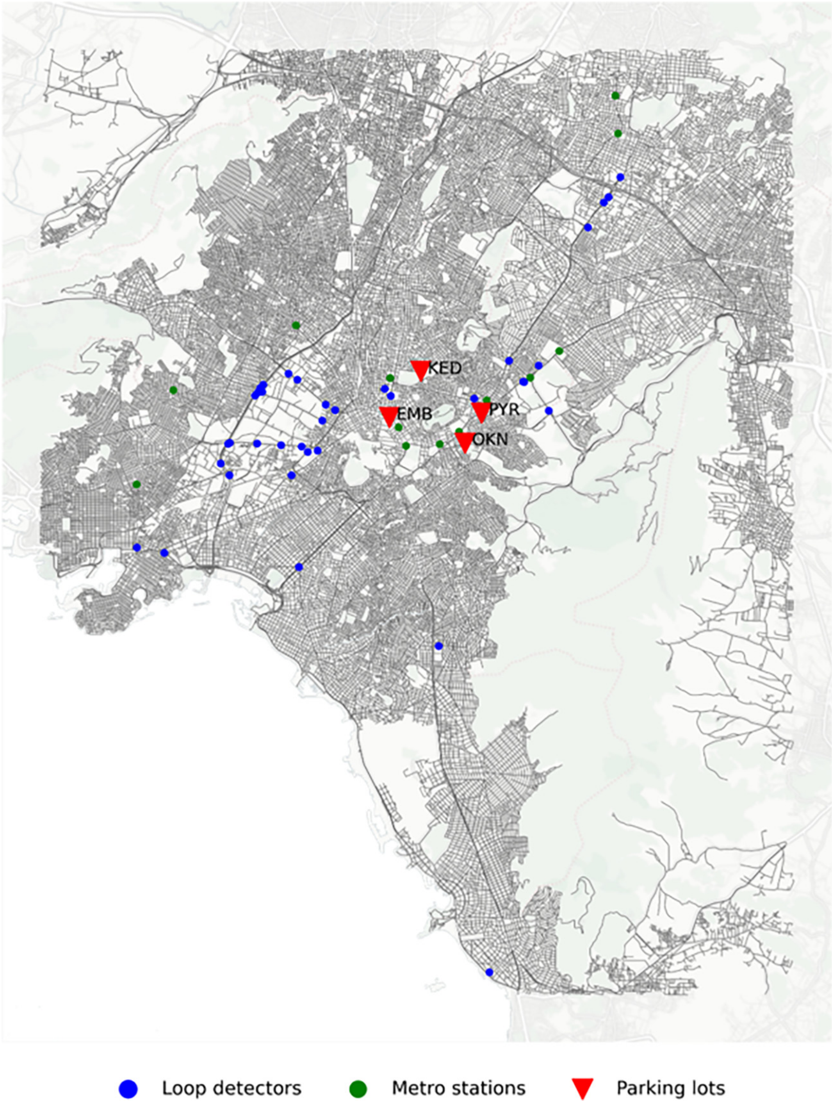

Fig. 7 Spatial distribution of the parking facilities and their most correlated loop detectors and metro stations.

Table 1
Hyperparameter values of the XGBoost model.

Learning rate	0.12
Number of estimators	70
Maximum depth	6
Minimun child weight	1
Column sample by tree	0.8
Subsample	0.9

series include temperature readings. Literature suggests that weather conditions play a significant role in parking decisions, and by incorporating these variables, the model can capture these effects. Also, the previous parking occupancies are used as input (Table 3).

For that analysis, 6 models were developed to forecast parking occupancy at different time horizons: 1 h, 3 h, 5 h, 8 h, 12 h, and 24 h ahead. These time horizons were chosen to investigate the potential of providing drivers with information about parking availability before even their trip starts. For the training, 80% of the sequences were used, while the remaining 20% are used for testing.

Table 2
Hyperparameter values of the LSTM with attention models.

Loss function	MSE
Optimizer	Adam
Training epochs	20
Batch size	16

Table 3
Description of variables used in the training processes.

Category	Variable	Type	Unit
Temporal	Hour of the day	Categorical	–
	Day of the week	Categorical	–
	Weekend	Binary	
Weather	Temperature	Numeric	°C
	Rain	Binary	–
Traffic	Traffic volumes	Numeric	Vehicles/lane
Transit	Ticket validations	Numeric	–
Parking	Previous parking occupancies	Numeric	–

The LSTM models were trained using backpropagation through time (BPTT). The loss function was chosen to be MSE. The Adam optimizer was used to update the model parameters. The training was performed using a batch size of 16, which means that the model weights were updated after processing every 16 samples from the training data. The model was trained for 20 epochs, with early stopping based on validation loss. These hyperparameters of the model were derived through grid search to find the optimal combination for the model's performance (Table 2).

4.2 Results

In the present section, the results of the proposed framework are presented and specifically, the error goodness of fit metrics of the developed models are presented alongside the feature importance and SHAP values of the input variables of the XGBoost model and the Attention weights analysis for the LSTMs.

4.2.1 Baseline methods

To support the use of multimodal data for parking predictions, an additional LSTM model using only historical parking occupancy data to predict 1 h ahead was created. This model had an MSE of 0.011 9, an MAE of 0.073 9, an R^2 of 0.844 6, and an MAPE of 17.53% (Table 4). While these results demonstrate that historical occupancy data alone can provide reasonably accurate short-term predictions, there is still room for improvement by incorporating multimodal data into the models. Therefore, no additional models based solely on historical parking data were developed.

Table 4
Error and goodness of fit metrics for all the developed models.

	LSTM model without multimodal data			"Reference day" model		XGBoost
MSE	0.011 9			0.026		0.003 6
MAPE	17.53%			40.45%		14.02%
MAE	0.073 9			0.089 3		0.035 4
R^2	0.844 6			0.633 5		0.936 1
LSTM with attention models						
	1 h	3 h	5 h	8 h	12 h	24 h
MSE	0.002 6	0.007 5	0.012 1	0.013 2	0.016	0.013 3
MAPE	**8.77**%	15.96%	16.88%	17.14%	22.86%	19.71%
MAE	0.037 6	0.066 9	0.073 2	0.077 8	0.089 6	0.080 4
R^2	0.965 3	0.900 8	0.840 5	0.824 7	0.789 6	0.826 1
Transformer models						
	1 h	3 h	5 h	8 h	12 h	24 h
MSE	0.002 7	0.005 5	0.005 7	0.026 3	0.011 8	0.006 8
MAPE	9.01%	10.14%	10.69%	17.12%	19.12%	13.72%
MAE	0.038 8	0.048 4	0.049 3	0.077 6	0.074 1	0.057 9
R^2	0.964 5	0.927 8	0.918 8	0.825 1	0.844 9	0.910 5

To further justify the effectiveness of using complex deep learning models, we compared the performance of our developed models to the simpler "reference day" model. This model assumes that parking occupancy at a given time can be predicted by using the occupancy observed at the same time one week prior. As shown in Table 4, the "reference day" model achieved an MAPE of 40.45%, indicating a considerable prediction error. This highlights the limitations of such a straightforward method and the potential advantage of using complex models that can capture temporal dependencies.

4.2.2 XGBoost performance

The performance of the XGBoost model indicates that the variety of the input variables exploited allows the estimation of parking occupancy rates relatively well. The MAPE of 14.02% is a decent result, that demonstrates the model's ability to capture key patterns and relationships within the multimodal dataset despite its non-sequential learning approach. The exact values of the error and goodness of fit metrics of the prediction models are presented in Table 4. In addition to the main XGBoost model, five additional models were developed to predict parking occupancy 3 h, 5 h, 8 h, 12 h, and 24 h ahead. However, these models exhibited higher error rates, with MAPE values exceeding 20%. As a result, these models are not further presented in this analysis.

Fig. 8 depicts the scatter plot of predicted and actual values. It is observed that most points are to some extent close to the regression line, indicating the good performance of the model. However, there are some deviations, particularly at higher occupancy values.

Fig. 9 demonstrates the top 10 features contributing to the performance of the model in predicting parking occupancy based on PI. The most influential features are primarily traffic-related, showcasing the critical role of traffic flow data in accurately forecasting parking occupancy. Notably, the most important features are derived from data recorded 1 to 4 h prior, indicating that short-term historical traffic patterns are the most predictive for parking demand. Additionally, temporal variables, such as day of the week, and weather variables, like temperature, are included in the top 10. However, their lower importance suggests that they provide useful but secondary context compared to traffic-related features.

Fig. 10 depicts the SHAP summary plot to show the impact of each of the most influencing features on the model's output. Similar to PI, the SHAP values emphasize the role of traffic-related features and the importance of recent traffic patterns. The SHAP values provide additional insights into how these features affect the model's predictions. Specifically, higher traffic flows at key loop detectors are associated with higher parking occupancy predictions, reflecting the natural relationship between increased vehicular movement and parking demand. Also, lower temperatures tend to increase parking occupancy predictions, possibly due to a preference for car use during colder weather. Finally, day of the week also shows a consistent effect, with weekdays showing increased occupancies.

A high value of PI means that the model's accuracy depends a lot on a specific variable, while high SHAP value indicates the specified variable has a significant effect on increasing or decreasing occupancy prediction. A comparative representation of the PI and the mean of the absolute SHAP values can be seen in Fig. 11. Beyond any doubt, the traffic volume from loop detector MS933, which is located on a major arterial road leading into the city center, with data from 3 h ago is particularly significant. This observation likely holds importance, because traffic flow on major arterials leading into the city center often reflects the volume of commuters heading to work, school, or other activities. The data from 3 h prior could capture the peak inbound traffic during this commuting window, which directly impacts parking demand at facilities near the city center. The 1 to 4 h lag represents the time it takes for a spike in traffic flow to translate into increased parking occupancy. The spatial distribution of the most important loop detectors reflects a balance of monitoring high-demand central locations and capturing the effects of incoming traffic from peripheral areas. An interesting observation is the inclusion of loop detector MS815, which is farther south, in the most important loop detectors. This may represent an entry point to the city for suburban commuters.

4.2.3 Sequential models performance

The performance of the sequential models, i.e., the LSTM with attention model and the transformer-based model, is evaluated across various time horizons: 1 h, 3 h, 5 h, 8 h, 12 h, and 24 h ahead. As expected, the performance of both models gradually declines as the prediction horizon extends. The models in larger time horizons cannot capture the finer details and fluctuations in the occupancy patterns, probably due to the accumulation of uncertainty. The only exception is that the predictions for 12 h ahead are slightly worse than those of 24 h ahead. One possible reason for that could be that the 24-hour predictions tend to resemble the overall patterns observed in shorter, more immediate time frames. In contrast, the 12-hour horizon falls into an intermediate range where the model might struggle to balance between capturing immediate short-term fluctuations and long-term patterns. Table 4 shows the exact values of error and goodness of fit metrics of the time-series forecasting models. The LSTM with attention exhibits stronger performance for shorter horizons, while the transformer-based model exhibits better stability, particularly in medium- and long-term predictions. The transformer model's self-attention mechanism, which is not restricted to sequential processing, allows it to focus on critical timesteps regardless of their position in the sequence. This flexibility helps the transformer handle non-linear dependencies and long-term patterns more effectively, which may explain its relatively strong performance for long-term forecasts.

Fig. 12 compares the actual and predicted occupancy time-series for the 1-hour horizon (best performance) and the 12-hour horizon (worst performance) across the four parking facilities. For the 1-hour horizon, the model demonstrates high accuracy, with the predicted values closely following the actual values, indicating that the model effectively captures short-term occupancy patterns. On the other hand, for the 12-hour horizon, the predictions deviate more from the actual

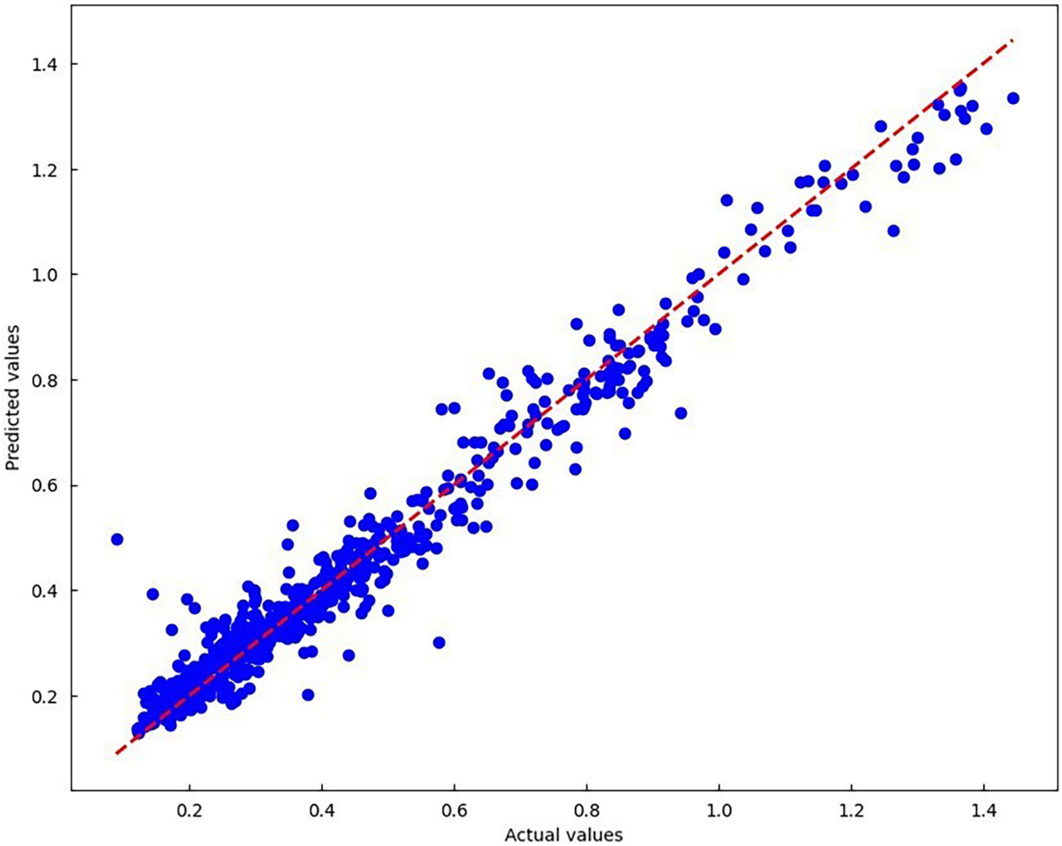

Fig. 8 Scatter plot of actual and predicted occupancies.

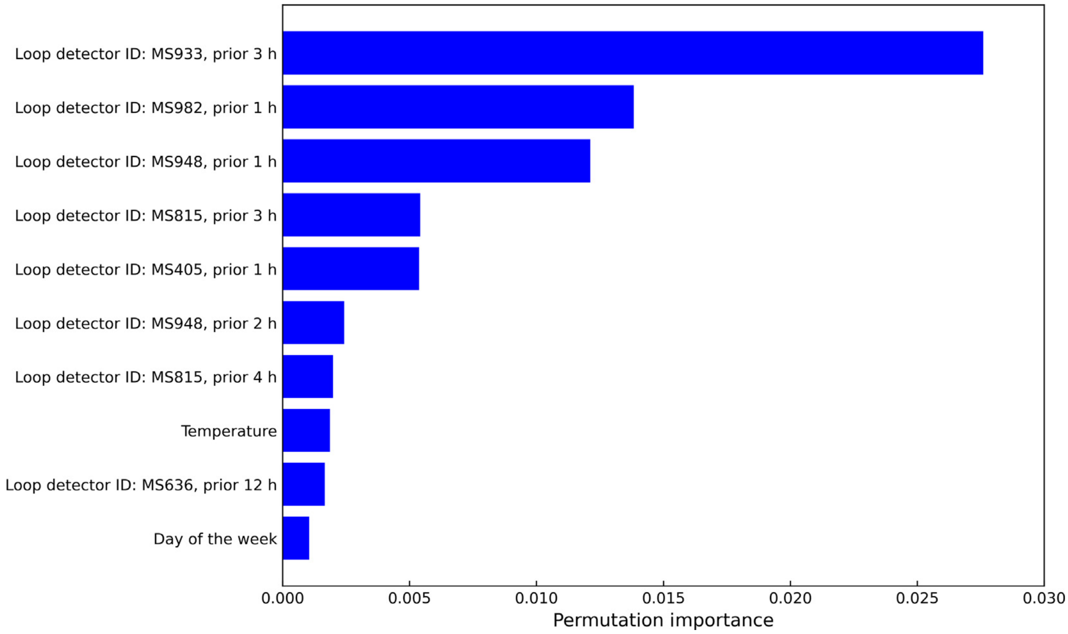

Fig. 9 PI of the independent variables.

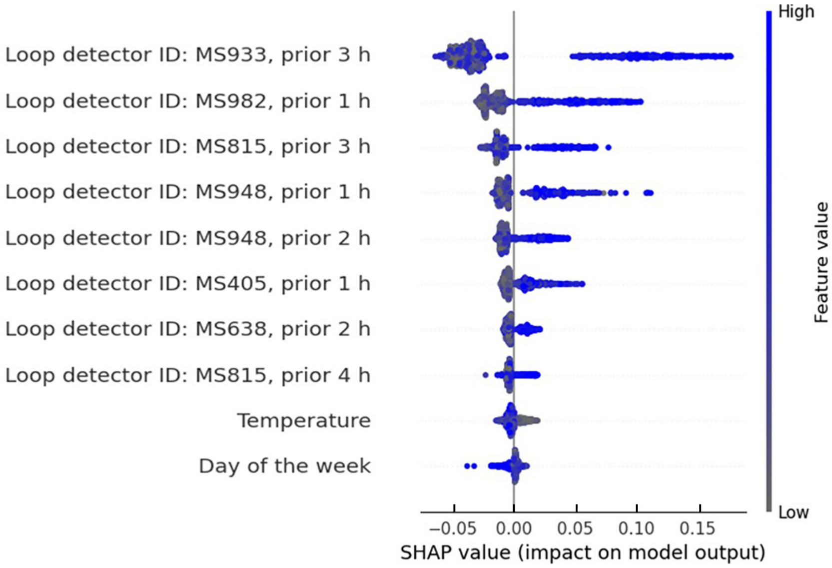

Fig. 10 SHAP summary plot with the XGBoost model.

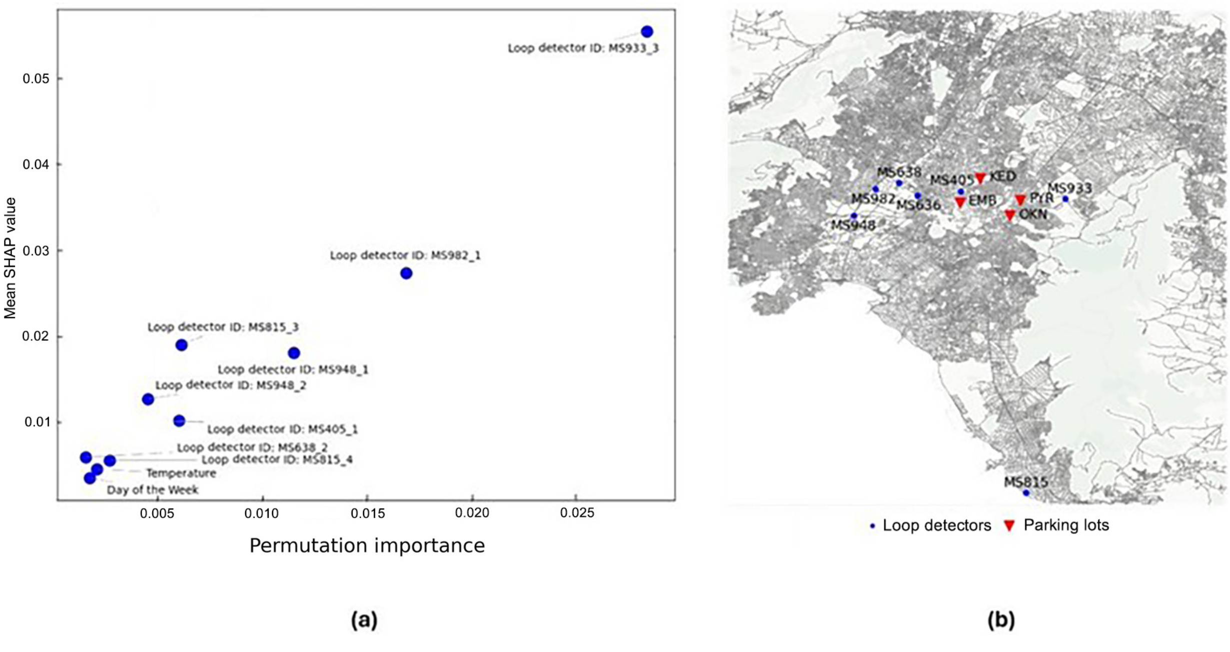

Fig. 11 PI and SHAP values of the independent variables (a) and map showing the spatial distribution of the most important loop detectors (b) for the XGBoost model.

values, highlighting the model's declining accuracy over longer time horizons. For the sake of brevity, the plots for the intermediate time horizons and the 24-hour horizon are not presented. However, these plots show the gradual decrease in the predictions' accuracy as the forecast horizon extends. Specifically, the predicted values deviate more noticeably from the actual values, with fluctuations and details in the occupancy patterns becoming less accurately captured.

Despite this general decline in accuracy over longer horizons, the error and goodness of fit metrics indicate that the models, especially the transformer-based, perform well in medium-term predictions, particularly for the 3-hour and 5-hour horizons. This is evidenced by relatively low MAPE and high R^2 values for these timeframes, suggesting that the model maintains a good balance between capturing necessary details and managing prediction uncertainty over these medium-term periods. This good performance in medium-term forecasts makes the model a valuable tool for proactive parking management strategies, allowing for more effective planning. Also, the 24-hours transformer model demostrates stability and effectiveness for

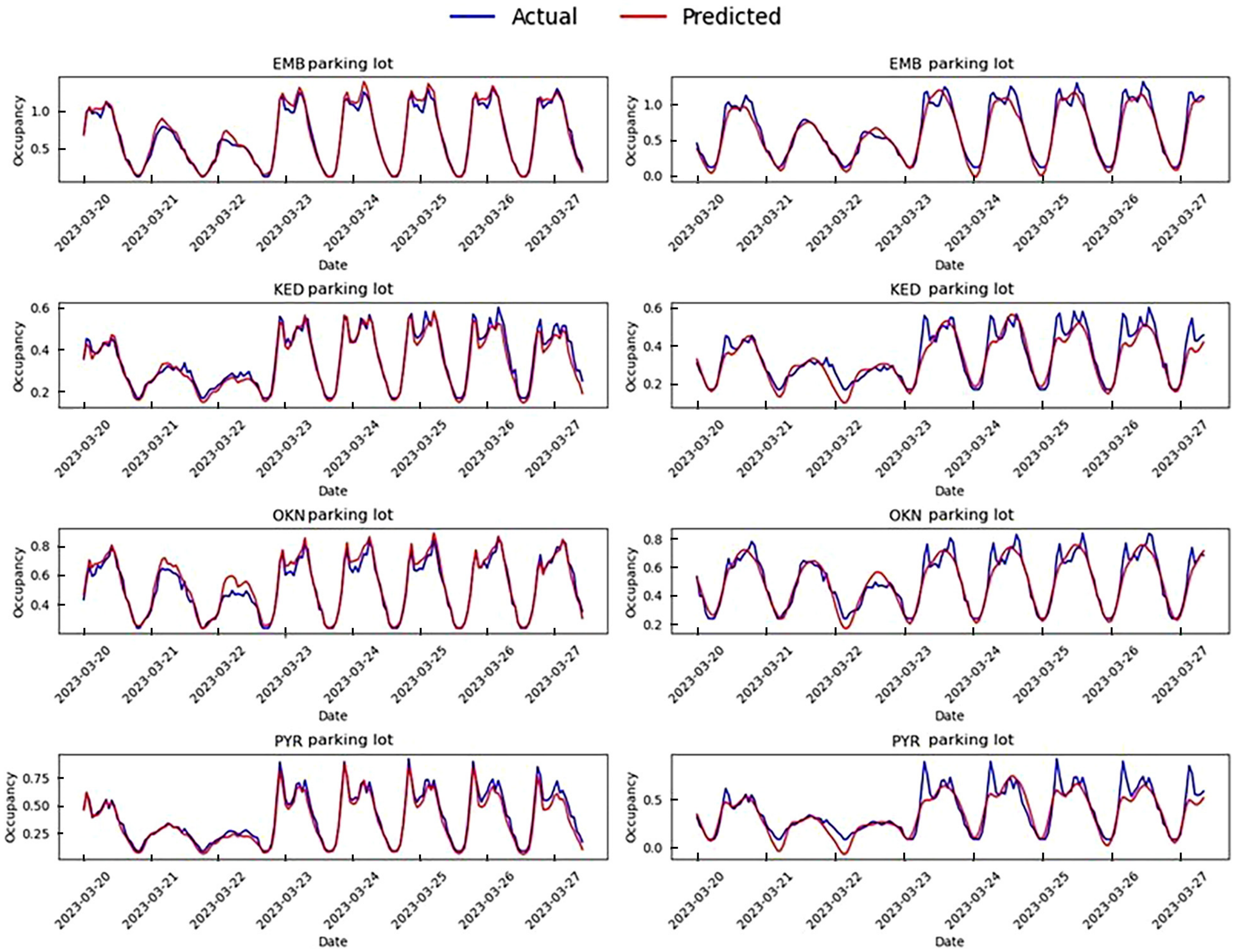

Fig. 12 Actual vs predicted occupancy timeseries for (a) 1-hour horizon, LSTM with attention model and (b) 12-hour horizon, transformer model.

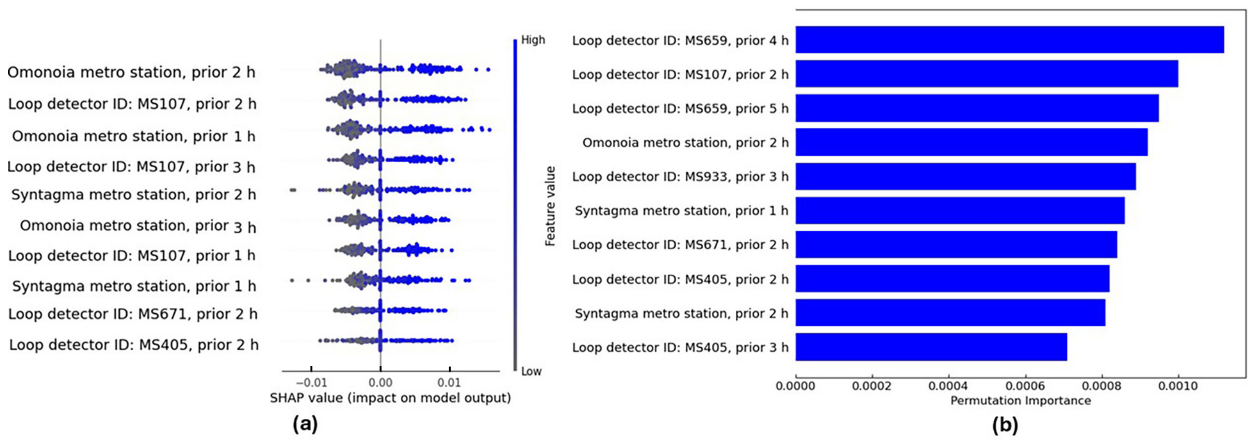

Fig. 13 SHAP summary plot (a) and PI (b) for the LSTM with attention model.

long-term forecasting. While there is a natural increase in MAPE and a slight decrease in R^2 compared to short-term and medium-term horizons, the transformer model's performance remains notably robust.

Additionally, from Fig. 12, the parking lots closer to the city center, i.e., the EMB and KED parking lots, tend to produce better prediction accuracy. This is mainly because these lots have more regular and consistent occupancy patterns, which the model can capture more effectively. Being in central locations, these lots experience more stable and predictable demand due to their proximity to major points of interest, business, and commercial districts. On the other hand, OKN parking, being the furthest from the city center and the main arterials, tends to show slightly lower prediction accuracy. This can be attributed to more fluctuating and less predictable occupancy patterns, likely influenced by the district's varying demand.

Comparing the two methods, the classic ML-approach with XGBoost achieves relatively high accuracy in predicting parking occupancy, but the sequential models have a better performance by capturing the temporal dependencies and incorporating variables that account for the entire transportation network. The only exception is the LSTM model that predicts 12 timesteps ahead. This model performs worse than the XGBoost model, with a higher MAPE (22.86%). The ability of LSTMs to forecast occupancy rates at various parking facilities and at various intervals enables the implementation of demand-responsive strategies and proactive information systems, which could help drivers and the management of the parking facilities.

For the explainability of the results, the SHAP values and PI for the best sequential model, i.e., the LSTM with an attention layer predicting 1 h ahead, were computed. From Fig. 13, it is clear that the LSTM model highlights the importance of not only loop detectors but also two of the most central metro stations in Athens, namely "Omonoia" and "Syntagma". Additionally, as expected, the SHAP values reveal that lower ticket validations or traffic volumes are associated with lower predicted parking occupancies. Furthermore, the analysis underscores the significance of recent past hours in influencing parking occupancy predictions, further emphasizing the temporal dependencies captured by the LSTM model. This aspect is further investigated in the following section, which focuses on the analysis of attention weights to better understand how the model prioritizes different time steps in its predictions.

Fig. 14 shows that ticket validations from Omonoia metro station and traffic volumes from loop detector MS107 from 2 h before are the most influential features across both explainability metrics. The key loop detectors identified are primarily located on major arterial roads leading to the city center, mirroring the findings of the XGBoost model. Interestingly, some of these loop detectors are consistently identified across both models, demonstrating a level of uniformity in feature importance. The consistency in feature importance across models demonstrates the robustness of these factors in explaining parking dynamics.

4.2.4 Attention weights

To compare the attention mechanisms of the LSTM and the transformer-based models, Fig. 15 visualizes the mean attention weights, i.e., the importance or relevance of each timestep of the input sequence for predicting a particular output, distributed over the 24 input timesteps for all the developed models.

For the LSTM model, shorter prediction horizons, such as the 1-hour and 3-hour models, the attention weights start at a higher level and decrease rapidly. This suggests that recent past hours are more important for predicting the near future. On the other hand, for medium-term prediction horizons, such as the 5-hour and 8-hour models, the attention weights start lower, decrease initially, and then slightly increase towards the end. This pattern indicates that both recent and some distant

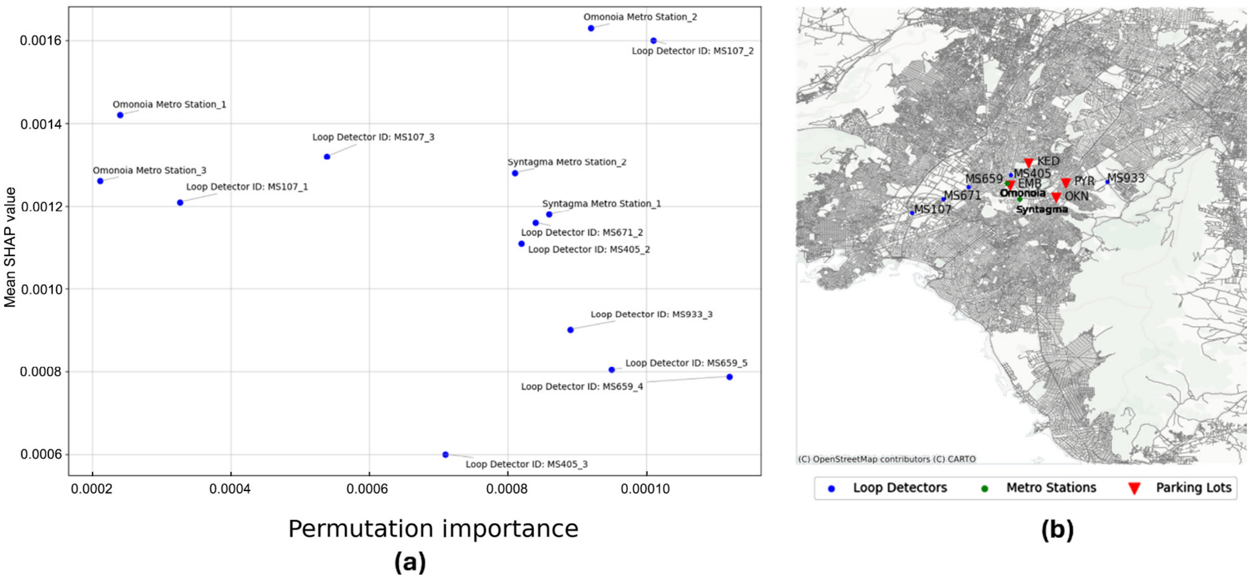

Fig. 14 PI and SHAP values of the independent variables (a) and map showing the spatial distribution of the most important loop detectors (b) for the LSTM model.

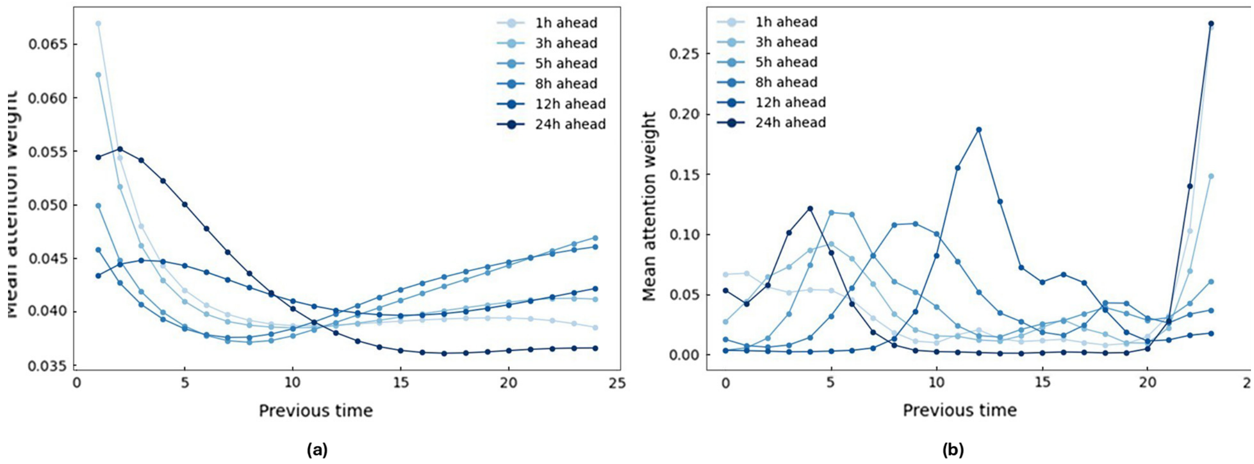

Fig. 15 Mean attention weights over previous time for different prediction horizons models.

past hours are important for these models. That can be explained as the model not only considers recent traffic conditions but also looks at traffic patterns from the same time of the previous day as they resemble the conditions of the timestep that it is trying to predict. For the 12-hour and 24-hour models, the attention weights exhibit a more complex pattern compared to shorter prediction horizons. Interestingly, the 12-hour ahead model assigns a relatively uniform importance to each of the past 24 h, indicating the models' need to balance immediate trends with long-term periodic patterns. The 24-hour ahead model weights indicate that the recent past hours are more important, because these hours likely reflect conditions similar to those being predicted.

In contrast, the transformer model displays a more varied distribution of attention weights across all prediction horizons. The shorter-term models (1-hour and 3-hour predictions) still emphasize recent hours, but the attention is more scattered, suggesting that the transformer model is capable of focusing on specific key moments rather than following a smooth pattern like the LSTM. For medium-term predictions (5-hour and 8-hour), the attention weights become more dynamic, assigning importance to different time points across the input sequence. The 12-hour and 24-hour models exhibit a complex pattern, with peaks at various points, indicating that the transformer model is not just focusing on the most recent data but is also identifying key time points throughout the 24-hour sequence that contribute significantly to the prediction.

5 Conclusions

The present research proposed accurate models for parking occupancy prediction and revealed comprehensive insights by analyzing parking data enhanced with traffic, transit, and weather data in two distinct levels. The first level focused on individual parking facility occupancy predictions using classic and explainable ML-techniques. The second level extended the analysis to model the relationships between the parking facilities occupancy and the broader transportation network using time-series forecasting and sequential deep learning models (LSTM and transformer), to investigate how many time-steps ahead predictions can be made to provide drivers with information regarding parking availability in advance. This includes an integration of traffic volumes and public transit ticket validations to account for a multimodal approach.

Results indicate that most sequential models outperform the XGBoost model in capturing the temporal dependencies. For shorter prediction horizons, such as the 1-hour and 3-hour models, the attention weights from the LSTM model prioritize recent past hours, emphasizing the importance of immediate trends in short-term forecasting. For medium-term predictions (5-hour and 8-hour horizons), the model balances recent data with patterns from the same hour on the previous day, demonstrating that both short-term fluctuations and long-term periodic patterns are critical. In contrast, the transformer-based model exhibited a more structured pattern, with various peaks. This suggests that the model effectively captures daily patterns, making it particularly well-suited for long-term predictions. The sequential model's, especially the transformer-based model's, ability to accurately forecast parking occupancy rates for the four off-street parking facilities and over different time horizons, especially in medium-term forecasting, makes it a powerful tool for implementing demand responsive strategies and providing real-time information to drivers. The SHAP value analysis for the LSTM model further highlights the significance of key features, such as recent traffic volumes and metro station ticket validations, in shaping short-term parking occupancy predictions, providing an interpretable framework for understanding the model's decision-making process across different prediction horizons.

Despite the XGBoost model's worse performance compared to the time-series approach, useful insights into the influential factors can be derived and used for decision-making from the parking facilities management. Specifically, the most critical features identified were traffic volumes from specific loop detectors, with lagged data from 1 to 4 h prior. Higher traffic volumes consistently corresponded to increased parking occupancies, possibly because greater vehicle volumes on arterial roads reflect a stream of commuters that increase demand for parking spaces, especially near central business districts and areas of high activity. Temperatures also play an important role, with lower temperatures increasing occupancy, possibly because colder weather often discourages people from walking or using alternative forms of transportation. Furthermore, drivers may prefer off-street parking during colder weather due to better protection, enhanced convenience, and proximity to points of interest. Finally, the day of the week variable highlights the reduced parking occupancy on weekends, likely due to more leisure-oriented travel and reduced commuting activity.

The multimodal approach that integrates traffic volumes, transit ticket validations, and weather data was demonstrated to significantly enhance model accuracy. The models with multimodal data outperformed the simpler model using only historical parking data, as well as the "reference day" baseline model. This highlights the limitations of basic prediction methods and the necessity of more sophisticated models that utilize multimodal data sources for improving occupancy forecasting.

The proposed framework can be used to inform users and enhance urban parking management strategies. Accurate predictions of parking occupancy enable the implementation of demand-responsive strategies such as dynamic pricing, which can adjust parking fees based on occupancy forecasting, ensuring optimal use of parking spaces and reducing congestion. Proactive information systems can be developed to guide drivers to facilities with available parking spots and incorporate price information. These models could also provide valuable insights for urban planners in designing and managing parking. The explainability of the results could help the decision-making to ensure optimal revenue and occupancy levels and prevent over-utilization or under-utilization of parking facilities.

The main limitation of this paper is that the current models focus only on off-street parking and do not include on-street parking data. Using on-street parking data could provide a more accurate view of parking dynamics and improve the performance of occupancy predictions. On-street parking is a significant component of urban parking supply, and should be also included in the predictions in future research. Also, the transferability of the models in different settings should be investigated.

Authors contributions

The authors confirm contribution to the paper as follows: study conception, design, analysis, interpretation of results and draft manuscript preparation: Katerina Vakrinou, Eleni G. Mantouka, Eleni I. Vlahogianni. All authors reviewed the results and approved the final version of the manuscript.

CRediT authorship contribution statement

Katerina Vakrinou: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Eleni G. Mantouka:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Eleni I. Vlahogianni:** Writing – review & editing, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Conflict of interest

Dr. Eleni Vlahogianni is an editorial board member/editor-in-chief for International Journal of Transportation Science and Technology and was not involved in the editorial review or the decision to publish this article. All authors declare that there are no competing interests.

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